



MA 612: Operator Theory

Course Code	MA 612
Course Name	Operator Theory
L-T-P-C	3-1-0-4
Prerequisites	MA 521 (Functional Analysis)
Intended for	M.Sc./B.Tech/M.S./M.Tech/Ph.D.
Distribution	Elective
Approval	31st BoA (Sep 2019); 23rd Senate (Oct 2019)

Course Contents

- **Elementary Spectral Theory** [18 lectures]
Banach Algebras, Examples of Banach Algebra, Spectrum and the Spectral Radius, Neumann Series, The Fundamental Theorem of Banach Algebra by Gelfand, Gelfand-Mazur Theorem, The Beurling Theorem for Spectral Radius, The Gelfand Representation, Compact and Fredholm Operators, Integral Operators, Kernels of the Integral Operators, Volterra Integral Operator, Transpose of the Bounded Linear Maps between Banach Spaces, Bounded Below Linear Maps between Banach Spaces, Ascent and Descent, Index of Bounded Linear Maps between Banach Spaces, The Fundamental Result of Fredholm Theory, Fredholm Alternative, Characterization of Fredholm Operators, Essential Spectrum in terms of Fredholm.
- **C*-Algebras and Hilbert Space Operators** [18 lectures]
Involution on an Algebra, *-Algebra, *-Algebra Generated by a Subset of *-Algebra, *-homomorphism between *-Algebras, Banach *-Algebras, Unital Banach *-Algebras, C*-Algebras, Examples of C*-Algebras, Double Centralizer for a C*-Algebra, Multiplier Algebra, Complete Characterization of Abelian C*-Algebras using the Gelfand Representation, Spectral Mapping Theorem, Positive Elements of C*-Algebras, Operators and Sesquilinear Forms, Adjoint and its properties, Orthogonal Projections, Invariant Subspaces, Reducing Subspaces, Partial Isometries and their Characterizations, Polar Decomposition, Compact Hilbert Space Operators, Diagonalizable Operators, Diagonalizability of Compact Normal Operators on Hilbert Spaces, denseness of the set of finite-rank operators, Hilbert-Schmidt Operators, Trace-Class Operators, The Spectral Theorem for Normal Operators.
- **Gelfand-Naimark Theorem** [6 lectures]
Ideals in C*-algebras, Positive Linear Functionals, Gelfand-Naimark-Segal Representation.

Text Books

1. Gerard J. Murphy, **C*-algebras and Operator Theory**, Academic Press, 2014.
2. Ronald G. Douglas, **Banach Algebra Techniques in Operator Theory**, Volume 179 of Graduate Texts in Mathematics, Springer, 2012.

References

1. Kenneth R. Davidson, **C*-algebras by Example**, American Mathematical Society, 1996.
2. Kehe Zhu, **An Introduction to Operator Algebras**, Volume 9 of Studies in Advanced Mathematics, CRC Press, 2018.
3. John B. Conway, **A Course in Operator Theory**, American Mathematical Society, 2000.